

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India 09/2021 – 01/2025 (3.5 Years)

- Developed and optimized ML compilers using **PyTorch** and **C++** for a novel NPU architecture, enhancing processing capabilities by **18%**.
- Designed and implemented advanced **quantization techniques** to improve ML model efficiency and accuracy, reducing model size by **30%**.
- Optimized **ML workloads** for deployment on diverse hardware, ensuring top-tier performance and energy efficiency for large-scale deployments.
- Collaborated with strategic customers to address specific ML challenges, delivering tailored solutions that improved system performance by an average of **22%**.
- Researched, prototyped, and implemented innovative solutions in **ML compiler design** and system-level optimizations, contributing to a new internal framework.
- Evaluated and debugged **ML performance** across complex workflows, identifying and resolving bottlenecks that reduced execution times by up to **25%**.

SKILLS

ML Development – PyTorch, C++, TensorFlow, ONNX, scikit-learn, R

Compiler & Optimization – ML Compilers, Quantization Techniques, Performance Optimization, System-Level Optimizations

Programming Languages – Python, C++, Java, SQL

Cloud & Infrastructure – AWS, Docker, Linux

Tools & Methodologies – Git, Jenkins, Jira, Agile/Scrum, CI/CD

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin 02/2026

BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India 01/2021

PROJECTS

NPU Architecture Compiler Optimization Framework

- Developed a compiler framework in **C++** and **PyTorch** to target a specialized NPU architecture, focusing on optimizing tensor operations.
- Implemented novel **quantization algorithms** to reduce model footprint and inference latency, achieving a **28%** reduction in memory usage.
- Integrated performance profiling tools to analyze ML workload execution, identifying and resolving bottlenecks that increased throughput by **15%**.

Energy-Efficient ML Workload Deployment Pipeline

- Engineered a Python-based pipeline for optimizing and deploying ML models, emphasizing energy efficiency for embedded systems.
- Designed a modular system for applying various **quantization and pruning techniques**, allowing for flexible trade-offs between accuracy and performance.
- Collaborated with a simulated customer team to test and validate optimized models on target hardware, gathering feedback for iterative improvements.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)